

1. Illustrate the operation of FIND-MAXIMUM-SUBARRAY on the array $A = \langle -5, 3, -2, 4, -7, -6, 1, -2 \rangle$. 4分

2. Use the master method to give tight asymptotic bounds for the following recurrences.

(a) $T(n) = 9T(n/3) + n$.

(b) $T(n) = 9T(n/3) + n^2$.

(c) $T(n) = 9T(n/3) + n^3$.

6分

1.

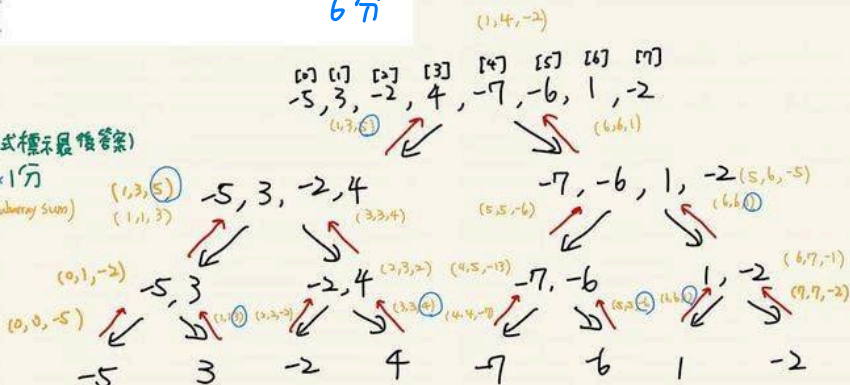
→ divide 1分

→ conquer (以任意形式標示最後答案)

0: left, cross, right of max 1分

(subarray low 位置, subarray high 位置, subarray sum)

2分 (至少寫出 sum)



max-subarray = 5

2. (a) $T(n) = 9T(\frac{n}{3}) + n \Rightarrow a=9, b=3, f(n)=n$

1分 $n^{\log_b a} = n^{\log_3 9} = n^2 > f(n)$, 且 $\epsilon = 2-1=1 > 0$

1分 $\therefore$ by case 1, $T(n) = \theta(n^2)$

(b) $T(n) = 9T(\frac{n}{3}) + n^2 \Rightarrow a=9, b=3, f(n)=n^2$

1分 $n^{\log_b a} = n^{\log_3 9} = n^2 = f(n)$

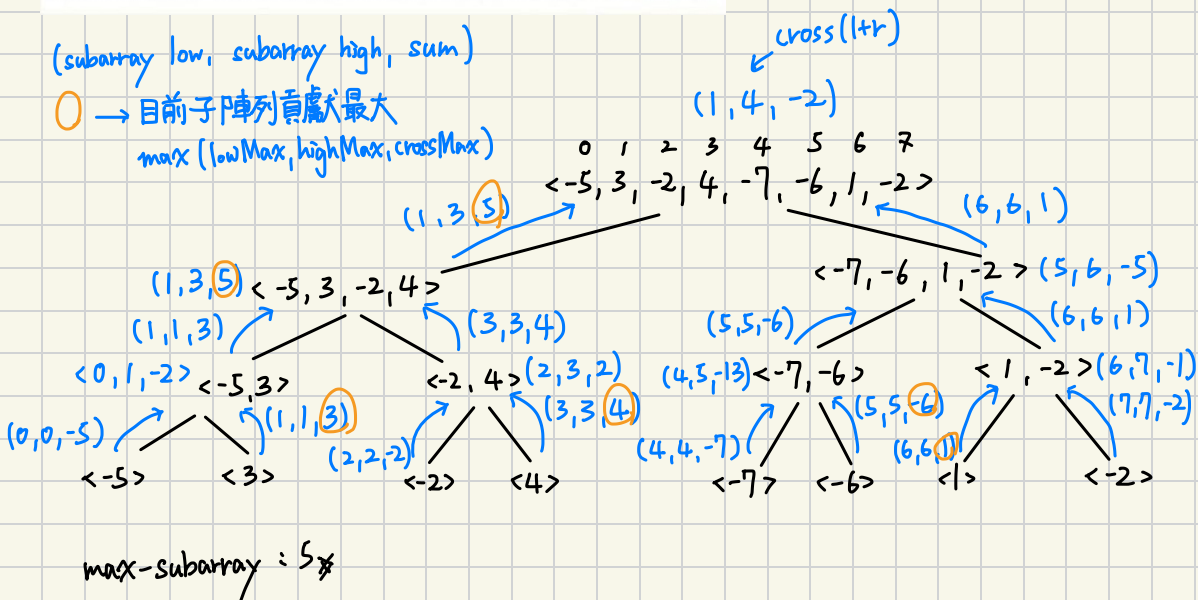
1分 $\therefore$ by case 2, $T(n) = \theta(n^2 \lg n)$

(c) $T(n) = 9T(\frac{n}{3}) + n^3 \Rightarrow a=9, b=3, f(n)=n^3$

1分 $n^{\log_b a} = n^{\log_3 9} = n^2 < f(n)$, 且 $\epsilon = 3-2=1 > 0$

1分 $\left\{ \begin{array}{l} a f(\frac{n}{b}) \leq c f(n) \Rightarrow 9 f(\frac{n}{3}) \leq c f(n) \\ \Rightarrow 9 (\frac{n}{3})^3 \leq c n^3 \quad (c < 1) \\ \Rightarrow c \geq \frac{1}{3} \text{ 滿足正數條件} \\ \therefore \text{成立, by case 3, } T(n) = \theta(n^3) \end{array} \right.$

1. Illustrate the operation of FIND-MAXIMUM-SUBARRAY on the array $A = \langle -5, 3, -2, 4, -7, -6, 1, -2 \rangle$.



2. Use the master method to give tight asymptotic bounds for the following recurrences.

(a) $T(n) = 9T(n/3) + n$

(b) $T(n) = 9T(n/3) + n^2$

(c) $T(n) = 9T(n/3) + n^3$

key: $T(n) = aT(\frac{n}{b}) + f(n)$

比較 $n^{\log_b a}$ 和 $f(n)$ 之成長速率。

Regularity Cond.

$af(\frac{n}{b}) \leq cf(n), c < 1$

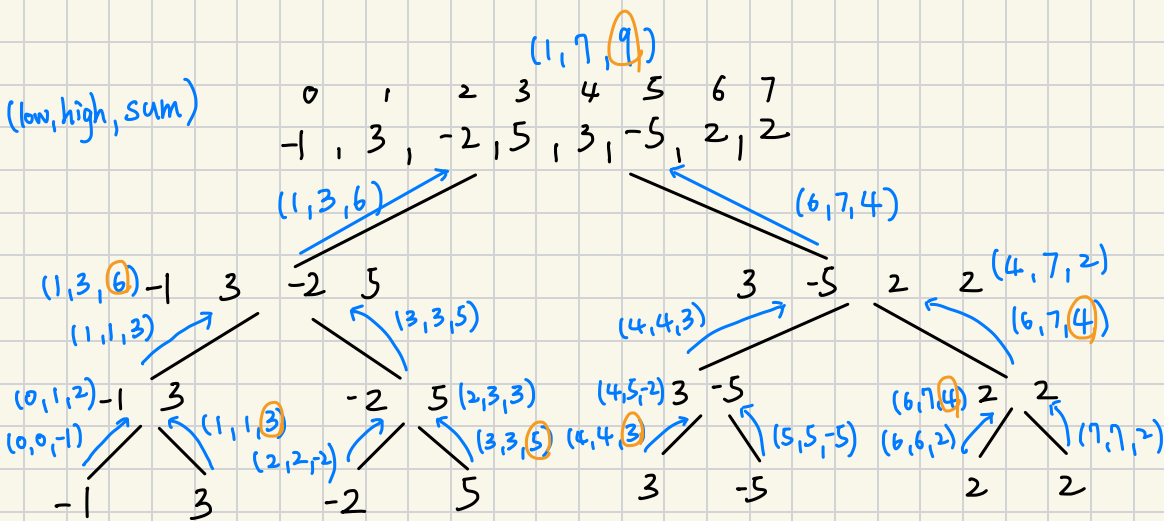
(a) $T(n) = 9T(\frac{n}{3}) + n, a=9, b=3, f(n)=n$
 $n^{\log_b a} = n^{\log_3 9} = n^2, n^2 > n$ 且 $\underline{\varepsilon = 2 - 1 = 1 > 0}$
 $\therefore$ by Case 1, $T(n) = \Theta(n^2)$

(b) $T(n) = 9T(\frac{n}{3}) + n^2, a=9, b=3, f(n)=n^2$
 $n^{\log_b a} = n^{\log_3 9} = n^2, n^2 = n^2$
 $\therefore$ by Case 2, $T(n) = \Theta(n^2 \lg n)$

(c) $T(n) = 9T(\frac{n}{3}) + n^3, a=9, b=3, f(n)=n^3$
 $n^{\log_b a} = n^{\log_3 9} = n^2 < n^3$ 且 $\underline{\varepsilon = 3 - 2 = 1 > 0}$
 $af(\frac{n}{b}) \leq cf(n) \rightarrow 9f(\frac{n}{3}) \leq cf(n) \rightarrow 9(\frac{n}{3})^3 \leq cn^3$
 $\rightarrow \frac{1}{3}n^3 \leq cn^3$

$\therefore$ ①、②成立, by Case 3, $T(n) = \Theta(n^3) \rightarrow \underline{c \geq \frac{1}{3}}$ 即滿足 Regularity Condition. ($c < 1$) ②

(low, high, sum)



max-subarray: 9

主定理 The Master Method

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

a 个子問題, 每个子問題大小 $\frac{n}{b}$

$T\left(\frac{n}{b}\right)$: 每个子問題所耗時間

$f(n)$: 分割合併所耗時間

3 種 Case (🧐: 比較 $n^{\log_b a}$ 與 $f(n)$ 的成長速度)

Case 1 $n^{\log_b a} > f(n)$ 相差 $n^\epsilon \rightarrow \theta(n^{\log_b a})$

Case 2 $n^{\log_b a} = f(n) \rightarrow \theta(n^{\log_b a} \lg n)$

Case 3 $n^{\log_b a} < f(n)$ 相差 n^ϵ 且滿足正規條件 $\rightarrow \theta(f(n))$

不適用於所有 $f(n)$ 。

當 $n^{\log_b a} = n$, $f(n) = n \log n$ 時, 雖然 $n < n \log n$, 不過大的程度不及 n^ϵ (一個多項式因子)。

此時, 遞迴式會落在 Case 2 和 Case 3 之間, 無法使用 Master Method。

正規條件 Regularity Condition: (PPT P.30)

* 確保隨遞迴深度增加, 子問題的總工作量會減少。

$$af\left(\frac{n}{b}\right) \leq cf(n), \quad \underline{c < 1}$$